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Dissolved organic carbon in a constructed and natural fens in the Athabasca oil sands region, Alberta, Canada

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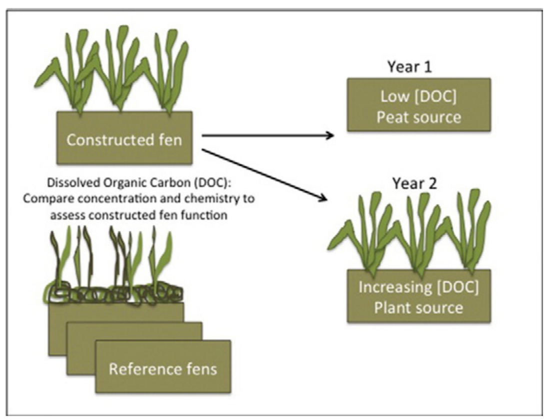
HIGHLIGHTS

- DOC concentration and chemistry were monitored in a constructed fen (CF)
- Concentration was lower in CF than reference fens, increasing more at CF over time
- DOC absorbance indicated larger, more humified character at CF than reference fens
- Temporal shifts in DOC absorbance at CF were opposite to patterns at reference fens
- Changes in DOC chemistry at constructed fen in year 2 suggest increasing plant input

GRAPHICAL ABSTRACT



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ABSTRACT

In the Athabasca oil sands region near Fort McMurray, Alberta, Canada, peatlands are disturbed extensively in order to recover bitumen below the surface. Hence, following oil sands mining, landscape reclamation is a part of the mine closure process in order to return functioning ecosystems, including peatlands, to the region. This study was conducted at a pilot fen reclamation project and three other diverse natural (poor, rich and saline) fens in the oil sands region during the growing seasons of 2013 and 2014, the first and second year post-construction. Ecosystem functioning of the constructed fen (CF) was evaluated with reference to natural fens based on pore water dissolved organic carbon (DOC) concentration and chemistry. Significant variation of DOC concentration among the reference fens was observed, varying from an average of 42.0 mg/L at the rich fen (RF) to 70.8 mg/L at the saline fen (SF). Dissolved organic carbon concentration at CF was significantly lower than at all reference fens, but increased significantly over the first two years. Seasonal variation of DOC concentration was also observed in each site with concentration increasing over the growing season. At CF, DOC was comprised of larger, more humic and complex aromatic compounds than reference fens in the first year post-construction based on its spectrophotometric properties; however, these differences were reduced in the second year. Initial DOC concentration and chemistry at CF was indicative of the source being largely the peat placed during fen construction. Changes in chemistry and increasing concentration of DOC in the second growing season likely resulted from increasing inputs from plants established on site. These results suggest that DOC

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concentration is likely to increase in future at CF as vascular plant productivity increases and in response to salinity sourced from tailing sand used to construct the catchment.

1. Introduction

In the Athabasca oil sands development areas near Fort McMurray, Alberta, Canada 65% of the pre-mining landscape was covered by fen peatlands (Vitt and Chee, 1989) with 895 km^2 cleared for surface mining (Government of Alberta, 2013). Surface mining removes large tracts of undisturbed boreal forest, including peatlands, to recover bitumen below the surface. Hence, reclamation is a part of the mine closure process required by the government of Alberta in an effort to recover natural ecosystem functions. Fen reclamation is a new concept and peatland construction design has not been extensively tested yet at the field-scale. One study by Amon et al. (2005) succeeded in establishing fen plants and a thin deposit of organic matter on small plots lined with gravel and fed by calcareous artesian water, while Vitt et al. (2011) reported successful establishment of fen plants on a former well pad used for oil sands extraction.

In 2010, a new approach for fen reclamation was initiated in the post oil sands landscape in order to validate fen model recommendations (Price et al., 2010). The goal of fen reclamation is to create a self-sustaining ecosystem that is carbon accumulating, capable of supporting peat- forming plant species, and resilient to normal periodic stresses (e.g., drought; Daly et al., 2012). A variety of revegetation strategies are being tested on the fen based on the moss layer transfer technique (Rochefort et al., 2003) and other North American fen restoration studies (Cooper and Macdonald, 2000; Vitt et al., 2011). Aquatic carbon fluxes and organic matter quality have both been defined as peatland ecosystem functional characteristics targeted by reclamation (Nwaishi et al., 2015a). The current study is focused on the evaluation of ecosystem functioning of the constructed fen (CF) compared to natural fens in the region based on dissolved organic carbon (DOC) dynamics.

Dissolved organic carbon is operationally defined as carbon contained in organic molecules that pass through a 0.45 μm filter (Thurman, 1985). Dissolved organic carbon is produced by partial decomposition of organic matter and subsequent dissolution in pore water and export of DOC is an important component in peatland carbon balances (e.g., Billett et al., 2004). High net production of DOC in peatlands is mainly due to slow microbial decomposition of plant material because of anoxic soil conditions (Moore and Dalva, 2001). Increased primary production can also result in higher DOC concentrations because exudates from plants contribute DOC (Freeman et al., 2004). In the boreal region, peatlands supply most of the DOC found in downstream ecosystems (Mulholland and Kuenzler, 1979; Urban et al., 1989; Dalva and Moore, 1991; Molot and Dillon, 1996), representing significant regional redistribution of terrestrial carbon. Moreover, the export of carbon from aquatic ecosystems to the atmosphere (i.e., release of CO_2 from exported DOC by microbial decomposition) may have profound influences on the regional carbon budget and the ecology of the region (Dalva and Moore, 1991; Schiff et al., 1998).

Dissolved organic carbon represents a relatively small fraction (0.04%–0.2%) of the total organic matter in soil (Zsolnay, 1996); however, it is often perceived as the most active fraction of soil organic matter due to its mobility and presumed labile nature (Zsolnay, 1996). According to Steinberg (2003), DOC is a complex mixture of organic compounds varying widely in molecular weight, from small molecules like organic acids and sugars to intermediate polymers like hemicellulose and large polycondensates usually called humic matter (including fulvic acids, humic acids and humin). Moore et al. (2003) determined that fulvic acid accounted for 58 to 97% of the total DOC in peatland pore water and contributed substantially to peatland acidity.

Spectroscopy in the ultraviolet and visible spectral ranges has been used for the characterization of DOC in peatlands. Absorbance ratios (E2/E3 = 250/365 nm and E4/E6 = 465/665 nm) and specific UV absorbance (SUVA) have been used to investigate DOC chemistry. A high value for E2/E3 ratio is indicative of low aromaticity and low molecular size of aquatic humic solutes (Grayson and Holden, 2012). The absorbance ratio E4/E6 indicates the fulvic vs. humic nature of DOC (Hautala et al., 2000; Spencer et al., 2007). A ratio < 5 and >6 indicates humic and fulvic acids, respectively, which in turn gives information about the molecular weight and aromaticity (Christopher et al., 2006). In addition, SUVA is determined by normalizing absorbance at 254 or 250 nm with DOC concentration and used to measure aromatic character of DOC in soils, with higher SUVA indicating higher aromaticity (Weishaar et al., 2003). Evaluating differences in absorbance ratios between the constructed fen and reference fens in the region may be useful in evaluating sources and processes contributing to net DOC production post-reclamation.

1.1. Controls on DOC concentration and export

A number of studies have reported increasing DOC export over the past several decades due to a rising air temperature resulting from climate change, roughly doubling DOC production with each 10 °C rise in temperature (Cole et al., 2002; Moore et al., 2008; Worrall et al., 2008). Higher temperature leads to greater microbial activity and enhanced decomposition of peat and thus, increased production and consumption of DOC. However, Lumsdon et al. (2005) have proposed a model of DOC runoff from high-organic matter soils based upon changes in DOC solubility. With increasing temperature, the DOC hydrophilicity is enhanced, leading to greater DOC solubility and export. Rising air temperature can also result in deeper peatland water table (WT) if evapotranspiration increases exceed precipitation increases (e.g., Roulet et al., 1992); this can lead to changes of DOC concentration, due to the exposure of different portions of the peat profile for aerobic and/or anaerobic decomposition. Higher DOC concentration is often reported upon rewetting of the peatland after a period of drought (Kalbitz et al., 2000) suggesting that WT fluctuation is an important control for net DOC production.

Further, hydrology is the major control on DOC distribution patterns, export and quality from natural peatland ecosystems (Buffam et al., 2001; Freeman et al., 2001; Waddington et al., 2008). This hydrology-DOC relationship is largely controlled by climatic variables affecting the water budget of the site, and peatland hydrologic characteristics such as WT position, hydraulic conductivity through the peat profile, hydraulic gradient across the peatland (Fraser et al., 2001) and the composition and distribution of microforms (Belyea and Clymo, 2001). Export of DOC is generally greater from peatlands with higher measured discharge (Fraser et al., 2001; Freeman et al., 2001). Fluxes of DOC in freshwater systems increase during storm events; some studies have shown that a large proportion of the annual DOC export (36 to >50%) occurs during short-duration high-intensity rainfall events (Hinton et al., 1997; Buffam et al., 2001).

Vegetation community, composition and productivity are also important for controlling pore water DOC concentration since the production of DOC from various litter types is different (Kuiters, 1993; Currie et al., 1996). Enhanced vegetation productivity is correlated with higher DOC concentrations in soil solution (Kang et al., 2001; Freeman et al., 2004). Moreover, the rate of carbon cycling can vary greatly between microforms within a peatland due to varying environmental conditions. Several studies suggest that carbon accumulation is higher at hummocks than hollows (e.g., Belyea and Clymo, 2001; Strack et al., 2006). However, these studies generally ignore hydrologic fluxes of carbon, particularly between microforms. It has been found that hummocks may lose carbon as DOC while hollows gain it (Strack et al., 2008).

1.2. Fen reclamation and DOC dynamics

The basic challenge for fen reclamation is to design a groundwater system that can support the inflow of water required to sustain hydrological and biogeochemical functions within the peatland and minimize the adverse effect of poor water quality derived from tailings sand commonly used in construction of post-mining landscapes. Tailings materials result in oil sand process-affected water (OSPW) including sodium (Na) and naphthenic acids (NAs), which can be toxic to wetland plants (Trites and Bayley, 2008; Apostol et al., 2004) and alter microbial communities (Nyman, 1999). In addition, elevated levels of salinity cause toxic accumulation of ions in plants, which inhibit plant growth (Jacobs and Timmer, 2005; Timmer and Teng, 2004). All of these adverse hydrological and biogeochemical conditions may affect the establishment of plant and microbial communities, ultimately limiting the DOC concentration in constructed peatlands. However, greenhouse experiments suggest that some fen species can survive exposure to OSPW (Federico et al., 2012; Pouliot et al., 2012).

This research aims to improve our understanding of the effect of fen reclamation on DOC dynamics. It also provides baseline data on DOC concentration and chemistry in natural fens useful for future DOC studies in the region and for the evaluation of reclamation success. Findings from the study can also help to improve management strategies to maximize carbon sequestration in reclaimed systems. The primary objectives of this study are to: 1) compare soil pore water DOC concentration and chemistry between the constructed fen (CF) and reference fens and 2) evaluate the environmental factors (pH, electrical conductivity (EC), temperature, and WT) affecting the concentration of DOC in CF and reference fens.

2. Study sites

2.1. Reference fens

Three diverse reference fens near Fort McMurray, Alberta were used in this study. They are referred to as poor fen (PF), rich fen (RF) and saline fen (SF) throughout this paper.

The poor fen is located ~40 km south of Fort McMurray (56° 22.610 N, 111°14.164 W). This fen features a forested upland surrounding the fen and also receives discharge from bogs within its boundaries. Plant species at the site included *Sphagnum* spp., *Chamaedaphne calyculata*, *Carex* spp., *Picea mariana* and *Betula pumila*. The PF basin was dominated by *Sphagnum* moss species (*Sphagnum fuscum* and *Sphagnum angustifolium*); however, distinct plant communities were observed in the north and south of the fen basin. The central part, with lower WT, had more trees than the northern and southern wetter parts, which were shrubbier. The peat depth was 4 m deep on average; however, thickness varied widely ranging from <1 m to >10 m. The fen was disturbed by a road at its west end leading to very wet conditions in this portion of the site.

The rich fen was located ~20 km north of Fort McMurray (56° 56.330 N, 111° 32.934 W). It was a treed moderate-rich fen. This site was disturbed due to cutlines and dirt roads passing through the broader fen boundaries although the actual study area has not been directly impacted. The site's vegetation was dominated by *Larix laricina*, *Betula pumila*, *Equisetum fluviatile*, *Smilcina trifolia*, *Carex* spp. and brown mosses, largely *Tomenthypnum nitens*. The peat is about 1 to 1.5 m thick.

The saline fen was located 10 km south of Fort McMurray (56° 34.398 N, 111° 16.518 W). This fen was dominated by *Juncus balticus*, *Calamagrostis stricta* and *Triglochin maritima*. It was an extremely saline site due to its geological setting that resulted in discharge of saline groundwater (Wells and Price, 2015). The site was surrounded by forested peatland but there were no trees in the study area. Peat depth was 0.75 to 1.5 m.

2.2. Constructed fen

The Nikanotee Fen is a constructed fen (CF) located ~30 km north of Fort McMurray (56° 55.8701 N, 111° 25.0166 W) built on a reclaimed area near Suncor's Millennium mine that was previously mined and later backfilled with overburden and lean oil sands. A low permeability geotextile liner was placed over the target site; petroleum coke forms the aquifer system, overlying the liner and extending partway up the catchment upland slope. The catchment was completed in fall 2012. At the lower end of the system, peat from a moderate-rich fen (2 m thick) was placed over the petroleum coke aquifer to form the fen in winter 2012–2013. The upland area, which contributes groundwater to the fen, was covered with boreal forest soils and was revegetated with shrubs and tree including *Larix laricina* and *Picea mariana* near the fen and *Pinus banksiana* at the uppermost part of the upland in fall 2013. The fen was designed based on recommendations from Price et al. (2010) such that water should infiltrate into the upland soils, percolate through the tailings sand, flow laterally through the petroleum coke aquifer, and discharge into the fen. As such, construction materials placed throughout the fen watershed could all potentially contribute DOC to the fen pore water. Moreover, salts, metals and petroleum contaminants in the petroleum coke and tailings sand can move along this flow path and affect water quality in the fen.

Within the fen, plant species (dominant in nearby peatlands) were introduced in the last week of June 2013. A variety of planting treatments were spread across the fen including moss transfer (Rochefort et al., 2003), planted sedge and shrub seedlings (including *Carex aquatilis*, *Juncus balticus*, *Betula glandulosa*, *Calamagrostis inexpansa*, *Triglochin maritima*), and seeding with native peatland species. The land surface was covered with wood mulch on half of all treatments so that loss of water by evaporation could be reduced. A spill-box was also installed in the fen basin in order to prevent flooding during snow melt and storm events. The total basin area is 32.1 ha, of which CF accounts for 2.9 ha.

3. Methods

3.1. Field instrumentation

In May 2012, three pairs of piezometer nests were installed in each reference fen, with one nest of the pair in a hummock and the other in a neighbouring hollow (six nests total at each fen). The pairs of piezometer nests were arranged in a transect that ran parallel to the dominant flow direction in each fen in order to capture variation along this flowpath. Each nest had piezometers at 50 cm, 75 cm and 100 cm depth, and a well. For CF, six piezometer nests were installed in two transects, each paralleling the main flow direction, on July 4, 2013 immediately after planting vegetation. Each nest had piezometers at 50 cm, 75 cm and 90 cm depth and a well.

Wells were constructed of PVC pipe (2.8 cm inner diameter) having 5 mm holes over the length of a 1 m long well. The pipe was wrapped from outside with a nylon stocking to prevent clogging. Each piezometer was slotted with 5 mm holes over a 17 cm intake that was installed so that it was centered at the desired sampling depth. There was a 100 mL reservoir below the piezometer intake to collect water samples. The slotted length of each piezometer was wrapped with 1.5 μm polypropylene mesh net (Nitex) to prevent movement of peat into the piezometer. A hand auger of the same diameter as the pipes was used to make holes through the peat at each site for installation of the wells and piezometers. Meteorological conditions (temperature and precipitation) for the study period were taken from the Environment Canada, Fort McMurray A meteorological station (http://climate.weather.gc.ca).

3.2. Water sampling

Piezometers were flushed and freshly recharged water was collected in a clean 60 mL Nalgene bottle after first rinsing it with a small amount of sample. The soil was wet enough, and hydraulic conductivity high enough, so that all the water samples could be collected on the same day as flushing. Environmental factors such as pH, temperature and EC were measured using a YSI-63 Multi-Parameter Probe (Xylem, Inc., Yellow Springs, OH, USA), and WT was recorded for each sampling location and date. Sampling was conducted every two weeks from May until September 2013.

In order to follow the development of the constructed fen over time, and based on the results from 2013, a reduced sampling regime was implemented in 2014. Following the procedure outlined above, 50 cm deep piezometers were sampled from each nest at each site monthly between May and August 2014. Access to RF and CF was limited in May 2014 due to wildlife activity and thus the first samples from these sites were collected in early June.

3.3. DOC concentration and chemistry analysis

All water samples collected from the field were pre-filtered the same day through a nominal 1.5 μm glass fibre filter (Sterlitech 934-AH), followed by filtration through a nominal 0.4 μm glass fibre filter (Sterlitech GB-140) and were transported in a cooler to the Physical Geography Laboratory at University of Calgary and stored at 4 °C until analysis.

All the water samples collected were analyzed using a PerkinElmer UV/VIS Spectrophotometer Lambda 35 to measure absorbance at 250 nm, 254 nm, 365 nm, 400 nm, 465 nm and 665 nm wavelengths. The absorbance ratios (E2/E3, E4/E6) and $SUVA_{254}$ were determined for DOC chemistry analysis. The absorbance by DOC of each sample was measured in three different ways: natural sample, diluted sample with deionized (DI) water (1:1 ratio) and neutralized samples using $NaHCO_3$. The absorbance ratios were calculated from neutralized samples (Kononova, 1966) whereas $SUVA_{254}$ was measured from natural samples (Weishaar et al., 2003). Diluted samples were measured in case high concentrations of DOC resulted in saturation of absorbance; however, as this was not observed, the values from the diluted samples were not used in this study.

A 20% subset of the total set of samples was randomly selected for Total Organic Carbon (TOC) analysis. These samples were diluted at 1:10 ratio in DI water and then acidified to lower the pH to ~2 using concentrated HCl. The concentration of DOC was determined using a Shimadzu TOC analyzer (Environmental Sciences Program, University of Calgary) by the Non-Purgeable Organic Carbon (NPOC) method. After evaluating all tested wavelengths for correlation to the measured DOC concentration, 254 nm was chosen as it best explained the variation in DOC concentration (see also Peacock et al., 2014). The regression equation ($y = 21.72x + 5.02$, $R^2 = 0.83$, $F_{1,\,119} = 491.51$, $p < 0.01$) was used to estimate DOC concentration in all samples.

3.4. Data analysis

Regression lines between DOC concentrations in 2013 and environmental variables such as pH, temperature, EC and WT and interaction of these variables with site were fitted using generalized estimating equation (GEE) repeated measures analysis of variance (ANOVA) to assess the strength and direction of the relationship and were considered significant if $p < 0.05$. All models were validated to ensure that assumptions were met (e.g., normality of residuals). In addition, GEE ANOVA was used to assess the differences of DOC concentration between sampling locations considering site, depth and microform and all two-way and three-way interactions as factors.

To evaluate changes in DOC concentration and chemistry over time, data from 50 cm depth in both 2013 and 2014 was considered. Since sampling started at CF only in early July of 2013 following the completion of construction and planting, samples included in the analysis were limited to those collected between late June and August in both years. GEE ANOVA was used to assess differences between years and sites. When significant differences were observed, pairwise comparisons with Bonferroni corrections were used to investigate significant differences between groups. All statistical analyses were performed using IBM SPSS statistics version 22.

4. Results

4.1. Environmental conditions

Total precipitation during the growing season (May–September) at the Fort McMurray A meteorological station was 296.6 mm and 307.4 mm for 2013 and 2014, respectively. Both years were wetter than the 30-year (1981–2010) climate normal of 283.4 mm over the same months. The distribution of precipitation was very different between the study years with over half of the seasonal precipitation arriving in June in 2013, largely during a storm on June 8 and 9, 2013 (Fig. 1). In contrast, precipitation was more evenly distributed among the months during 2014. Mean growing season air temperature was warmer in 2013 at 15.4 °C. Mean air temperature in 2014 was 13.8 °C, similar to the 30-year normal of 13.3 °C during the same months.

There was a variation of WT position in each site over both growing seasons (Fig. 1). However, the WT position across all sites was generally above the surface to within 20 cm below the surface throughout the study period. Heavy rains in mid-June 2013 resulted in elevated WT at all reference fens during the late-June sample date, including flooded conditions at the hollows at SF. Variation in WT at the reference fens resulted largely from the difference in WT between hummocks and hollows, whereas at CF it resulted from variable topography related to construction and settling of placed peat. On average, WT was slightly deeper in 2013 than 2014 except at SF (Table 1).

4.2. Pore water DOC concentration

There was no significant effect of microform type or depth on DOC concentration and no significant interaction between these factors with each other or with site (Table 2). Thus, these categorical variables were omitted from the following analysis. The concentration of DOC in soil pore water varied across the sites with individual samples ranging from 16.5 to 89.2 mg/L. The mean (± standard deviation) concentration between late June and August, considering both study seasons, at 50 cm depth at SF was the highest at 70.4 ± 9.8 mg/L followed by PF at 47.6 ± 11.6 mg/L and RF at 38.4 ± 9.7 mg/L (Fig. 2). Mean DOC concentration at 50 cm depth between July and August 2013 at CF was 30.4 ± 5.1 mg/L. In 2014, 50 cm deep pore water DOC concentration was 71.5 ± 7.6, 50.8 ± 9.2, 44.4 ± 7.2 and 39.8 ± 3.8 mg/L at SF, PF, RF and CF, respectively. Considering both years, 50 cm DOC concentration over this period was significantly different at each of the sites (Table 3; GEE, $\chi^2 = 385.1$, $p < 0.001$) with CF being the lowest. There were also significantly higher concentrations in 2014 compared to 2013 (GEE, $\chi^2 = 7.8$, $p = 0.005$). There was a significant interaction between site and year (GEE, $\chi^2 = 20.8$, $p < 0.001$) in which the concentration at CF and RF was significantly higher in 2014, while there was no significant difference between years at PF and SF. Despite the greater rate of increase in DOC concentration at CF between 2013 and 2014, DOC concentration remained significantly lower at CF in 2014 than all reference fens.

All of the sites exhibited temporal variation of DOC concentration throughout the growing season starting from May until September (Fig. 1). DOC concentration (50 cm depth) was the lowest in May, increasing to the highest value at the end of the growing season with a similar pattern in both years. Unlike the reference fens, at CF there

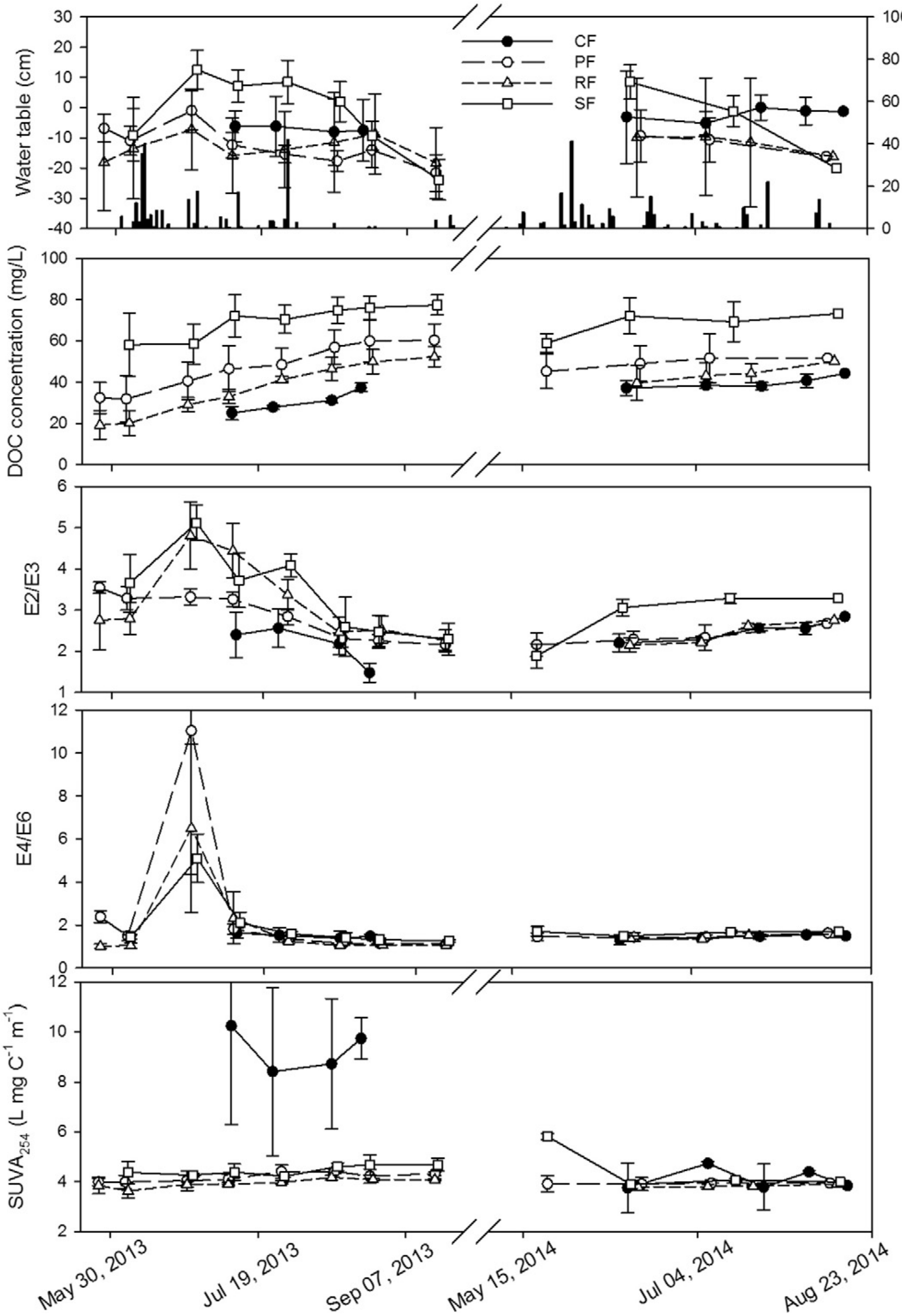

Fig. 1. Temporal variation of mean a) water table (WT) position b) DOC concentration, c) E2/E3, d) E4/E6 and e) $SUVA_{254}$ at each site. All water samples for DOC analysis were collected from 50 cm depth. The mean was calculated from six sampling locations from each site on each sampling day. Error bars show the standard deviation. Negative values for WT indicate depth below the surface. Daily precipitation measured at Fort McMurray A meteorological station is also shown as bars in a). CF is constructed fen, PF is poor fen, RF is rich fen, and SF is saline fen.

was no clear decline from the last fall measurement in 2013 to the first spring measurement in 2014.

4.3. Controls on pore water DOC concentration

Using the larger 2013 data set, the influence of pH, temperature, EC and WT was evaluated. There was a significant interaction between site and pH, EC, WT and temperature (Table 4). Across all sites, there was a significant positive correlation between EC and DOC concentration (Fig. 3); this was also the case at PF and RF, but no significant relationship was observed at CF and SF. DOC concentration generally decreased across all samples as pH increased, although this was not significant; the opposite pattern was observed at PF, RF, and SF but not CF, resulting in the significant site-pH interaction. Higher water temperature resulted in higher DOC concentration, but this was only significant at CF, PF and RF. Water table position did not significantly contribute to variation

Table 1
Hydrochemical conditions at the study sites.[a]

	Water table (cm)	pH	Corr conductivity[b] (μS cm^{-1})	50 cm depth water temperature (°C)
Poor fen				
2013	−13.7 (7.0)	4.7 (0.4)	27.9 (32.0)	13.9 (2.0)
2014	−12.1 (3.7)	5.0 (0.6)	76.4 (118)	n.m.
Rich fen				
2013	−12.6 (4.2)	6.8 (0.9)	312 (161)	12.7 (2.7)
2014	−11.8 (3.1)	7.7 (0.2)	405 (232)	n.m.
Saline fen				
2013	−0.5 (13.7)	6.5 (0.4)	13,800 (4150)	16.2 (3.1)
2014	−4.2 (14.5)	7.3 (0.2)	11,800 (2210)	n.m.
Constructed fen				
2013	−7.0 (0.9)	7.6 (0.2)	1830 (635)	12.4 (2.9)
2014	−2.1 (2.1)	7.7 (0.1)	2030 (646)	n.m.

[a] All values are mean (standard deviation) measured on samples from 50 cm depth (except water table), between late June and August. Water temperature was not measured (n.m.) in 2014.
[b] Corrected to account for conductivity of H^+ ions according to Sjors (1950).

in DOC concentration, except at PF where DOC concentration was higher when WT was deeper.

4.4. DOC chemistry

There was a common pattern of change for E2/E3 at all reference sites over the growing season with a lower value at the beginning of 2013 (Fig. 1), which increased to nearly double by the end of June, following a large precipitation event, and then continued to decline for the remainder of the summer. In 2014, E2/E3 gradually increased from May to September. CF followed a similar pattern of declining E2/E3 in 2013 and increasing values over the 2014 growing season; since sampling started in early July, response to the June 2013 storm was not observed at CF. This large event in June 2013 also resulted in a rapid shift in E4/E6 across all reference fens, with the highest value observed at PF. Values generally recovered by the next sampling date two weeks later and continued to decline until sampling ended in September. In 2014, E4/E6 increased gradually over the growing season at all reference fens and CF. $SUVA_{254}$ increased slightly over the growing season in 2013 at all reference fens and had little change in 2014, except for SF that had a high initial value in May. At CF, $SUVA_{254}$ values were very high in 2013, dropping to the range of values measured at reference fens by June 2014.

As with DOC concentration, site, year and site-year interaction were significant for explaining variation in E2/E3, E4/E6 and $SUVA_{254}$ of DOC at 50 cm between late June and August during the study years (Table 3). In 2013, CF had significantly lower E2/E3 than all the reference fens, while PF had significantly lower E2/E3 compared to RF and SF, which were similar. In 2014 E2/E3 of DOC at CF was similar to PF and RF. All these three sites had significantly higher E2/E3 than SF in 2014 (Fig. 4). For E4/E6, CF had a significantly lower value than all the reference fens in 2013, while the reference fens were not statistically

Table 2
GEE result showing the impact of microforms, depths and sites on DOC concentration.

Source	Type III		
	Wald χ^2	df	p
Intercept	5650	1	<0.01
Microforms	0.48	1	0.49
Depth	2.82	2	0.24
Site	544	2	<0.01
Microforms ∗ site	0.74	2	0.69
Microforms ∗ depth	1.13	2	0.57
Site ∗ depth	7.64	4	0.11
Microforms ∗ site ∗ depth	3.20	4	0.52

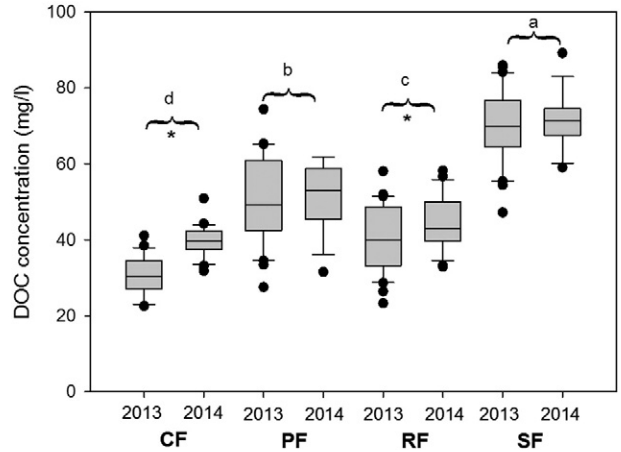

Fig. 2. Boxplots comparing DOC concentration at 50 cm depth between late June and August at each study site in 2013 and 2014. Study sites are significantly different (GEE ANOVA, $p < 0.05$) if they do not have letters in common. Stars indicate significant difference between the years at a given study site. The horizontal line in the middle of the box represents the median. Minimum and maximum whiskers represent the 10th and 90th percentiles, whereas lower and upper end of the boxes are first and third quartiles of the data, respectively. CF is constructed fen, PF is poor fen, RF is rich fen, and SF is saline fen.

significantly different from each other. In 2014, the E4/E6 of DOC at CF was significantly lower than SF, but not PF or RF (Fig. 4). Constructed fen had significantly higher $SUVA_{254}$ than all the reference fens in 2013. In 2014, $SUVA_{254}$ was statistically similar across all study sites.

5. Discussion

The constructed fen (CF) was designed with the goal to create a self-sustaining ecosystem that is carbon accumulating, capable of supporting peat-forming plant species, and resilient to normal periodic stresses (Daly et al., 2012). In evaluating the success of the design, a number of diverse natural fens within the same region were assessed for comparison to CF. However, the first two years following construction represents a very short duration to create typical peatland conditions. Therefore, the present study helps to evaluate the initial ecosystem functioning of CF but cannot explain how DOC concentrations and chemistry will change as the system develops in future.

5.1. Controls on DOC dynamics at the reference fens

The range of DOC concentration (10–89 mg/L) measured in pore water from the reference fens was within the concentration range observed in other peatlands (McKnight et al., 1985; Moore, 1987; Fraser et al., 2001; Kane et al., 2010). Both concentration and chemistry of DOC varied between years at the reference fens (Figs. 2, 4), likely due to differences in weather. This illustrates the natural variation in DOC dynamics that exists and illustrates the importance of continued monitoring of reference ecosystems during evaluation of reclamation success (e.g., Nwaishi et al., 2015a). As the difference in DOC concentration at CF was greater than that at the reference sites, and shifts in chemistry at CF were often opposite of those observed at the reference fens (e.g. E2/E3, Fig. 4), these changes at CF are more likely related to ecosystem development than shifts in allogenic controls between the seasons.

The rate of carbon cycling can vary greatly between microforms within a peatland due to varying environmental conditions. According to Strack et al. (2008) hummocks tend to lose carbon as DOC, while hollows/pools gain DOC; however, in the current study microtopography had little impact on DOC concentration. Similarly, DOC concentrations at depths 50 cm, 75 cm and 100 cm did not exhibit any significant differences. This is similar to the finding of a previous study carried out in northern boreal fens (Wyatt et al., 2012). Small changes in depth

Table 3
GEE result for site and year effect on variation in DOC concentration and chemistry at 50 cm depth between late June and August.

	df	DOC		E2/E3		E4/E6		$SUVA_{254}$	
		Wald χ^2	p	Wald χ^2	p	Wald χ^2	p	Wald χ^2	p
Site	3	385.1	<0.001	190.4	<0.001	55.3	<0.001	93.2	<0.001
Year	1	7.8	0.005	29.6	<0.001	31.2	<0.001	67.8	<0.001
Site * year	3	20.8	<0.001	50.1	<0.001	37.1	<0.001	54.8	<0.001
Intercept	1	5162	<0.001	10,102	<0.001	741.3	<0.001	2162	<0.001

below the subsurface may not have had an impact on DOC production because WT was close to the ground surface (within 20 cm) throughout the season in all sites, likely creating similar hydrochemical conditions within these shallow depths. In contrast, Moore et al. (2003) observed a decrease of mean pore water DOC concentration with depth ranging from 20 to 60 mg/L. Similarly, Höll et al. (2009) found significantly different DOC concentrations even at soil depths of 40 and 60 cm in similar water saturation conditions; however, these horizons differed significantly in the redox conditions, which may not be the case in the present study. Moreover, hillslope position can influence DOC concentration in peatlands (Boothroyd et al., 2015). This likely plays a minor role in the present study given the very low gradients present in the study sites. Transects were positioned parallel to the dominant flow direction at each site; therefore, any variation related to position along this flowpath is included in the reported mean values.

The concentration and chemistry of DOC varied among the reference fens. The production of DOC itself can alter local pore water chemistry, likely decreasing pH and increasing EC (e.g., Khadka et al., 2015). In general, this data set suggests that DOC concentration will be higher at sites with higher EC and warmer water temperature. Based on the pattern within each study site our data suggests that within a site, higher pH tends to be coincident with higher DOC concentration, while the pattern observed across sites is more likely driven by broad ecohydrological differences in site conditions as opposed to pH alone. The rich fen (RF) had the lowest mean DOC concentration, likely linked to its hydrochemistry and ecology. The shallow topography of the watershed around RF suggests that DOC was produced mainly within the system, which was rich with a variety of plant species. Freeman et al. (2004) state that higher primary production leads to higher DOC concentrations. However, lower DOC concentration observed at RF may be attributed to the possibility of photolysis and microbial uptake of DOC. Kelley et al. (1997) reported increased microbial activity in nutrient rich peatlands compared to poor sites suggesting that RF may have a microbial community that more efficiently consumes DOC. Moreover, RF had near-neutral pH, conditions resulting in higher DOC solubility (Fenner et al., 2009), which likely contributed to greater DOC export along with discharge water. Based on E2/E3, E4/E6 and $SUVA_{254}$ of the pore water DOC, RF had DOC that was of smaller molecular size and less aromatic than the other reference fens, factors that could lead to easier microbial degradation (e.g., Khadka et al., 2015) and greater mobility (e.g., Strack et al., 2011).

Table 4
GEE model results for environmental controls on DOC concentration.

Source	df	Wald χ^2	p
Temperature	1	0.049	0.83
Water table	1	2.58	0.11
pH	1	3.24	0.072
Electrical conductivity	1	27.7	<0.001
Site	1	18.0	<0.001
Site × temperature	3	10.8	0.013
Site × water table	3	8.60	0.035
Site × pH	3	30.0	<0.001
Site × electrical cond.	3	34.5	<0.001
Intercept	1	16.8	<0.001

Higher DOC concentration at PF was likely due to input from its watershed and biogeochemical conditions within the peatland. The poor fen was surrounded by steep upland slopes with shallow organic soils. The runoff from these uplands likely contributed both water and DOC to the fen basin. Koprivnjak and Moore (1992) found that DOC concentration in peatlands generally increased with increasing catchment size. These catchment contributions would also explain the substantial change in E4/E6 of DOC measured at PF following the large precipitation event in June 2013, as mobile DOC fractions produced in the uplands were rapidly flushed to the peatland. In addition, dominance of *Sphagnum* moss and high WT probably reduced the efficiency of decomposition resulting in more DOC production (Moore and Dalva, 2001). Furthermore, peat depth at the PF site, especially on the north side was nearly 10 times thicker than any other reference fens, which provided more peat volume to contribute to DOC production (Sachse et al., 2001).

The highest DOC concentration was observed at SF likely due to its extremely high salt concentration, neutral pH, comparatively warmer water temperatures, shallow WT, and vegetation dominated by sedges - optimum conditions for DOC production. We observed a positive correlation between EC and DOC concentration across all sites. Higher ion concentrations may result in higher DOC concentration by saturating ion exchange sites in the peat resulting in more DOC in solution (Kalbitz et al., 2000). Enhanced biodegradation, possibly leading to higher rates of net DOC production has been observed in the presence of salts due to flocculation, as larger structures will provide better attachment for microbial colonies (Marschner and Kalbitz, 2003). In contrast, Mavi et al. (2012) suggest that salinity hinders microbial activity, reducing the consumption of DOC and resulting in higher concentrations. Khadka et al. (2015) observed higher net DOC production in laboratory incubations of substrates under saline conditions across all study sites, providing additional evidence that high EC at SF was an important driver of the high DOC concentration. Moreover, the absence of trees at SF likely contributed to warmer pore water temperature

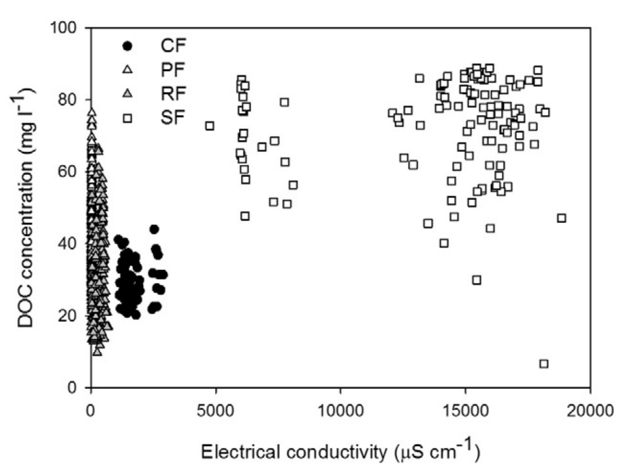

Fig. 3. DOC concentration versus electrical conductivity of pore water. Analysis based on the larger 2013 dataset including samples from 50, 75 and 100 cm depth. CF is constructed fen, PF is poor fen, RF is rich fen, and SF is saline fen.

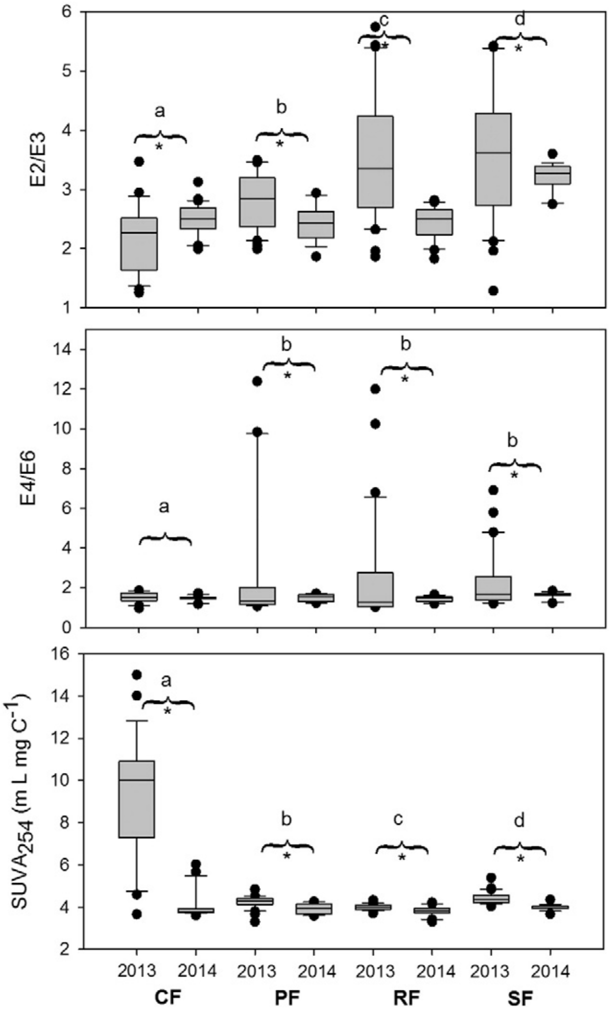

Fig. 4. Boxplots comparing a) E2/E3, b) E4/E6 and c) $SUVA_{254}$ of pore water DOC at 50 cm depth between late June and August at each study site in 2013 and 2014. Study sites are significantly different (GEE ANOVA, $p < 0.05$) if they have no letters in common. Stars indicate significant difference between the years at a given study site. The horizontal line in the middle of the box represents the median. Minimum and maximum whiskers represent the 10th and 90th percentiles, whereas lower and upper end of the boxes are first and third quartiles of the data, respectively. CF is constructed fen, PF is poor fen, RF is rich fen, and SF is saline fen.

(Table 1), which is a favourable condition for microbial growth, and higher solubility and net production of DOC (e.g., Clark et al., 2012; Khadka et al., 2015). In addition, litter from sedges is easier to decompose in comparison to moss and tree litters (Marschner and Kalbitz, 2003), and this could also lead to great net DOC production. The E2/E3 of DOC at SF suggests smaller size molecules than the other reference fens consistent with fresh plant residues or microbial cleavage of DOC (Steinberg, 2003), while E4/E6 and $SUVA_{254}$ values indicate more humic and aromatic content that may result from the increased solubility of these structures due to the saline conditions.

Hydrologic conditions, including larger growing season WT range, at SF may have also contributed to the higher DOC concentration. Increased DOC concentration occurring after the drop of the WT has been explained by the oxidation of peat material (Blodau et al., 2004) or by the enzymatic 'latch' mechanism (Freeman et al., 2001). Oxidative enzymes that are normally repressed in anoxic (wet) conditions become active in oxic (dry) conditions leading to the degradation of organic matter. DOC produced during dry periods can then be dissolved in soil water as the WT rises during wet periods. Furthermore, there was no regular channel for water discharge from SF, which may increase water residence time in soils and slow the loss of DOC.

Overall, production, absorption and consumption of DOC are factors within peatlands that could contribute to the differences in DOC concentration and chemistry among the sites. These factors along with the water pathways through the peatlands impact DOC concentration and chemistry (Moore and Dalva, 2001). However, the critical factors in determining DOC production rates are still not well understood because isolating variables in field conditions is very difficult (Moore and Dalva, 2001). Even though the sites were hydrologically and ecologically diverse and DOC concentrations were quite different, the seasonal patterns of DOC concentration and chemistry changes were similar among the sites across both years, suggesting a broader control on seasonal patterns linked to regional weather (i.e., air temperature and WT shifts driven by timing of warming and precipitation events). Seasonal patterns of pore water DOC concentration were consistent with previous studies (e.g. Scott et al., 1998; Strack et al., 2008; Kane et al., 2014) increasing over the growing season and declining again in the following spring. Over the growing season, shifts in E2/E3 to higher values suggest that the DOC pool has increasingly smaller molecules, while E4/E6 and $SUVA_{254}$ suggest slightly more humic and aromatic content. This likely indicates a plant contribution of low molecular weight humic substances (Kane et al., 2014) and microbial processing of DOC (Urban et al., 1989; Strack et al., 2008). The exception to this was the rapid shift in E2/E3 and E4/E6 in response to a large precipitation event in June 2013 (Fig. 1). These chemical changes suggest that low molecular weight, fulvic acids were mobilized during the event and subsequently made a substantial contribution to the peatland DOC pool. Similar patterns of shifts in E4/E6 for DOC exported from peatlands during storm events have been reported elsewhere (e.g., Grayson and Holden, 2012).

5.2. Constructed fen DOC dynamics: evaluating ecosystem function

The constructed fen had the lowest DOC concentration among the sites, although a significant increase in DOC concentration was observed between the first and second year post-construction (Fig. 2). In 2013, the first year post-construction, DOC was made up of significantly larger molecules with more humic and aromatic nature than any of the reference fens based on E2/E3, E4/E6 and $SUVA_{254}$. This is consistent with DOC sourced primarily from the placed peat. Prior to use in construction, this peat was dewatered for over two years and then dug up, moved, placed and compacted at CF; this process altered its physical and chemical properties, reducing organic matter quality (Nwaishi et al., 2015b) and resulting in lower rates of DOC production from this placed peat than peat collected from the reference fens (Khadka et al., 2015). DOC produced from peat also had lower E2/E3 and higher $SUVA_{254}$ than DOC produced from vascular plant litter (Khadka et al., 2015), suggesting that the pore water DOC measured in 2013 had little contribution from vegetation. In addition, construction materials used in the constructed fen's catchment (petroleum coke, tailings sand and upland soil) were poor sources of DOC (<1 mg/L per gram of dry weight as reported in Khadka et al., 2015). Furthermore, contamination of the fen by organic acids derived from tailings materials, especially naphthenic acids, likely contributed little to near surface measured DOC concentration given that measured concentration was <1 mg/L at 90 cm depth in 2013 and with no evidence of naphthenic acids at 50 cm even through the growing season of 2014 (Kessel and Price, 2015).

Vegetation type and productivity has also been associated with differences in DOC concentration and chemistry in peatlands. Strack et al. (2011) reported a positive correlation with DOC concentration and gross productivity at plots in a restored bog in Quebec. Recent photosynthates can contribute to the soil DOC pool at revegetated areas of extracted peatland (Trinder et al., 2008). Armstrong et al. (2012) report higher DOC concentration under the shrub *Calluna* than sedges or *Sphagnum* in UK peatlands, suggesting that plant type is also important. As plants were in the early establishment phase at the start of the present study, we did not specifically target our sampling to specific plant types; therefore, the effect of plant type on DOC dynamics at CF requires further study.

Despite many potential stresses on plants that may exist in constructed wetlands (e.g., Pouliot et al., 2012), successful establishment and growth of vascular plants at CF in 2013 resulted in large amounts of biomass accumulation by 2014 (data not shown). Litter deposition following growth in 2013 and continued expansion of plants in 2014 provided an additional potential source for DOC that would contribute, small, less humic and less aromatic DOC to the pore water pool (Khadka et al., 2015). Therefore, we hypothesize that this plant contribution resulted in the significant increase in DOC concentration at CF observed in 2014 as this was concurrent with a change in DOC chemistry that resulted in E2/E3 and $SUVA_{254}$ that were more similar to the reference fens (Fig. 4). This shift in DOC concentration and chemistry suggests that CF is functioning more similarly to natural fens in the region in terms of DOC dynamics after only two growing seasons. However, the present study did not specifically investigate the direct sources of DOC, its microbial processing or its export from the system, and these need to be better understood and monitored over time to fully assess functioning of CF in terms of net DOC production and export.

The impacts of environmental factors such as temperature, pH, EC and WT on DOC concentration were also investigated. Of these, EC had a profound impact on DOC concentration, which was exhibited by high concentrations at SF. Salinity likely provided nutrients to microbes and enhanced the net production of DOC. Moreover, Khadka et al. (2015) observed higher rates of DOC production from substrates collected from all the study sites in water with similar ion concentrations as CF, when compared to DOC production from the same substrates in distilled water. Higher EC in water at CF than RF (>5 times) could have played a role, to some extent, to compensate for the lack of DOC from fresh vegetation litter in 2013 at the former. As tailings sand used in construction of the upland slope at CF has a large potential source of salinity, increasing salinity over time at CF, in combination with high levels of plant productivity, could result in very high DOC concentrations (as observed at SF). Continued monitoring is required to assess this possible trend. Additionally, the concentration of specific ions and acid neutralizing capacity could play a more important role in DOC dynamics than EC alone (e.g., Clark et al., 2012) and chemistry at CF is likely to differ from the reference fens given the influence of OSPW. For example, availability of sulphate at CF is substantially higher than any of the reference fens (Wood et al., 2015; K. Murray, unpublished data). The role of specific ions on DOC concentration and chemistry at CF requires further study.

6. Implications for fen construction in the post-oil sands landscape

Following initial construction of a fen on an area previously disturbed by surface mining of oil sands, DOC concentration was significantly lower than three diverse reference fens studied in the region. The DOC in the first season post-construction also consisted of larger molecules with more humic, aromatic nature than that of the reference fens, based on E2/E3, E4/E6 and $SUVA_{254}$ of the DOC, likely due to reduced organic matter quality of the placed peat and limited plant productivity in the first growing season. The concentration and chemistry of DOC changed significantly at CF in the second growing season with chemistry indicative of increased input from plant litters and exudates. While the concentration remained significantly lower than all the reference fens, chemistry was more similar and not significantly different in many cases. As soil pore water DOC concentration and chemistry results from the combination of plant productivity, microbial processing of organic matter, pore water chemistry controls on DOC solubility and hydrological mobilization, and since it appears to respond quite quickly to changing conditions in the constructed fen, it is a useful indicator of ecosystem function (see also Nwaishi et al., 2015a). Spectrophotometric indicators of DOC quality are quick and inexpensive to measure, and provide some insight into controls on net DOC production and mobilization. Further investigation of DOC chemistry methods that more specifically describe the structure of the DOC (e.g., fluorescence spectroscopy, proton nuclear magnetic resonance spectroscopy, among others) may allow better separation of sources contributing to DOC pools and export (e.g., Tfaily et al., 2013).

As DOC concentration was initially low at CF, loss of carbon from the system as DOC is expected to be minor; however, total discharge of water from the system is also an important control on DOC export (e.g., Fraser et al., 2001; Waddington et al., 2008). The DOC concentration increased in the second growing season, likely due to rapid plant establishment, and we expect continued productivity will lead to even higher concentrations in the future. However, increasing concentrations of naphthenic acids and sodium over time, sourced from tailings sand used to construct the fen catchment, will likely lead to plant stress and the impact of this on DOC dynamics in the longer term is unclear. Moreover, higher EC resulted in higher DOC concentration suggesting that increasing salinity at CF over time could further enhance DOC production and loss. Therefore, as the system develops, DOC export could become a substantial portion of the carbon balance, depending on discharge, and its export should be monitored when evaluating carbon exchange at the site.

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